Sahil Singh

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Research Interests

Incoming PhD researcher in Engineering with a background in Applied Mathematics, specialising in computational modelling of physical systems. Experienced in solving PDEs, heat transfer problems, and implementing numerical methods (e.g. finite elements/difference) in Python. Focused on building reliable simulation workflows and applying mathematical models to optimise real-world engineering systems, particularly in fluid flow and thermal analysis.

Education

University of Nottingham Sep 2021 – Jun 2025
Integrated MMath in Applied Mathematics : First-class average (74%) *Nottingham, UK*
Award: Outstanding Community Contribution
Recognised for peer mentoring and positive impact on the student community; the panel highlighted the exceptionally high number of nominations received.

Selected Modules:

- **Multivariate Analysis (99%)** : PCA, clustering, CCA, factor analysis, multivariate regression; ranked **1st** in cohort.
- **Mathematics Dissertation (87%)** : Developed exact integration methods for physics-informed neural networks to solve PDEs with mesh-free training; ranked **2nd** in cohort.
- **Computational Applied Mathematics & Numerical Analysis (84%)** : finite elements, QR, low-rank solvers for high-dimensional PDEs and sparse systems; ranked **2nd** in cohort.
- **Optimisation (84%)** : convexity, KKT conditions, gradient methods, constrained optimisation.
- **Scientific Computation & Numerical Analysis (80%)** : stability/consistency, error analysis, iterative solvers.
- **Statistical Machine Learning (72%)** : team lead; regularised models, model selection, reproducible pipeline in **R**.

Technical Skills

Languages: Python, C++, R, Bash || **ML:** NumPy, SciPy, Matplotlib, scikit-learn, JAX, PyTorch, TensorFlow
Tools: Git || **OS:** Linux (Ubuntu)

Projects & Experience

Deep Learning Algorithms for Solving PDEs | *Python (NumPy/SciPy), C++, Matplotlib* Oct 2024 – Present

- Developed quadrature-free PDE solvers using analytic loss formulations, outperforming Monte Carlo methods.
- Implemented analytic assembly (mass matrix & projection vector) for shallow ReLU networks in 2D by computing areas/lengths of polygon intersections, eliminating sampling variance.
- Explored a Poisson solver on the unit triangle and explored the time-dependent heat equation; compared against FEM with a Monte Carlo baselines, documenting convergence behaviour and optimisation stability.
- Designed an alternating training scheme: exact outer coefficients via linear solve and inner ReLU parameters via BFGS with line search; added early stopping and reproducible experiment scripts.
- Produced evaluation tooling (error surfaces, ablations, plots) and unit-tested geometry kernels; wrote vectorised Python with small C++ prototypes for performance-critical polygon routines.

UN Development Indicators — Challenge | *R, tidyverse* Mar 2025 – Apr 2025

- Compressed a 70-year, 141-country panel to two PCs retaining 92% variance and visualised continental trends.
- Achieved 75% LOOCV continent classification with **PCA**$\rightarrow$**LDA**, ranking top 3 in cohort.
- Predicted 2007 life expectancy via 3-component PCR, beating a baseline by 17%.
- Delivered a fully reproducible **R Markdown** report with clear EDA and model-building workflow.

Chemical-Impurity Classification Challenge (Group Project) | *Python (scikit-learn, XGBoost), Git* Jan 2025 – Apr 2025

- Worked in a 4-member team to build end-to-end pipelines for two tasks: (i) ECG condition classification, (ii) impurity *type* (classification) and *percentage* (regression) from 5-channel spectra $+$ temperature.
- Led the classification stream: set up LDA/QDA and tree baselines, ran EDA (pairs/box plots, PCA/LDA), and co-designed a **hierarchical** route (single-variable LDA splits $\rightarrow$ LDA/SVM/RF branches) that outperformed simple models.
- Co-implemented the regression stack: OLS with interactions, Ridge/LASSO/Elastic Net with k-fold CV, plus ensembles (Random Forest, XGBoost); integrated classifier outputs into the regressor for joint scoring.
- Owned reproducibility: data cleaning/outlier handling, scaling, grid search, early stopping (where applicable), and Git workflows with peer reviews; co-authored the report and presented results.

Optimised Quadrature Rule Synthesis for Polygonal Domains | *C++, Custom Linear Algebra Modules* Nov 2024 – Dec 2024

- Engineered a full numerical pipeline to generate high-order, optimised quadrature rules with near-minimal points for arbitrary 2D polygonal domains decomposed into triangular elements.
- Implemented a recursive Newton-Raphson solver for roots of Legendre polynomials to construct 1D Gauss quadrature, extended to 2D on a reference square and triangle via tensor products and geometric mappings.
- Designed and implemented a non-negative least squares (NNLS) active-set optimisation algorithm to solve the constrained system $Ax = b, x_i \geq 0$; strategically removed points and recalculated weights to minimise rule size while preserving exact polynomial integration.
- Validated the system on complex domains (L-shapes, hexagons) against analytical solutions, demonstrating correct integration and the convergence properties of the optimised rules.

Researcher — Tracing the Hydrological Cycle Jul 2024 – Sep 2024

British Geological Survey & University of Nottingham *Hybrid*

- Assisted with field sampling; collected 20+ water-isotope samples for internal BGS use.
- Integrated 20-year flow and isotope series with tracer ODE models; refit parameters to improve validation fit across monitored sites.
- Delivered reproducible **R** notebooks and a short technical note on climate–isotope links.

Advanced Iterative Solvers & Eigenvalue Algorithms | *Python (NumPy, Matplotlib)* May 2024

- Implemented and benchmarked **Krylov subspace methods**: Steepest Descent (SD) and Conjugate Gradient (CG), for solving symmetric positive definite systems, evaluating convergence via residual norm plots.
- Extended CG to non-symmetric systems with the **CGNE algorithm**, solving normal equations $A^T Ax = A^T b$ and confirming convergence where standard CG fails.
- Developed **eigenvalue solvers**, including Power Iteration for dominant eigenvalues and Rayleigh Quotient Iteration for cubic convergence, analysing performance through residuals $\|\lambda v - Av\|_2$.
- Engineered a cost-efficient variant of the Power method that updates only the p most significant components per iteration, reducing computation while retaining convergence for large-scale problems.

Finite Element Methods & Numerical Linear Algebra Solver Suite | *Python (NumPy, Matplotlib)* Mar 2024

- Designed and implemented a full **Finite Element Method (FEM)** solver for 1D and 2D boundary-value problems, supporting Galerkin and Petrov–Galerkin formulations under Dirichlet and Neumann conditions.
- Extended the solver to assemble stiffness and mass matrices for second-order elliptic PDEs and **advection–diffusion** equations ($-\alpha u'' + \beta u' = f$), analysing stability and convergence across varying α, β regimes.
- Applied FEM semi-discretisation to the **heat equation**, combining Euler's method in time with Galerkin FEM in space to simulate temperature evolution.
- Built and benchmarked three **QR factorisation** algorithms (Classical, Modified, and Re-orthogonalised Gram–Schmidt), validating numerical stability on ill-conditioned test cases such as the Hilbert matrix.

Advanced Optimisation: Orthogonal Regression & Geographic Median | *Python (NumPy, Matplotlib)* Dec 2023

- Formulated high-dimensional **Orthogonal Regression** as minimising the sum of squared orthogonal distances to a hyperplane, deriving the closed-form solution via the minimum eigenvalue of the covariance matrix.
- Implemented the **Gradient Method** on the **Rayleigh quotient** to compute the dominant eigenvector, and proved theoretical convergence guarantees (validity of constant step size, non-vanishing iterates).
- Modelled a real-world **Fermat–Weber (geometric median)** problem to minimise weekly travel costs by weighting anchor locations by visit frequency, and solved it with the **Weiszfeld algorithm**.
- Delivered concrete results: an orthogonal regression fit with final objective value $f(x^*)$, and an apartment location optimising weekly transport costs under budget.

Optimisation Methods for Regression & Model Fitting | *Python (NumPy, pandas, Matplotlib)* Nov 2023

- Developed and derived explicit gradients and Hessians for core optimisation algorithms (Gradient Descent, Newton's Method, Gauss–Newton) to solve multivariate regression and non-linear least squares problems.
- Examined convergence behaviour by varying learning rates, showing stability for $t \leq 1.9$ and fastest convergence at $t = 0.9$, connecting empirical results to Lipschitz continuity.
- Extended Gauss–Newton with a damping strategy to fit a sinusoidal model to meteorological data, designing an adaptive step size $t_k = \min\{1,\ 1/\|\nabla g(x_k)\|\}$ to guarantee convergence to a physically realistic solution.
- Explored sensitivity of non-convex problems to initialisation, demonstrating convergence to distinct local minima and providing a theoretical rationale for the discrepancies in model parameters.